

QUESTION BANK
CHAPTER 13-NUCLEI
SECTION-A

MCQ

- Q1. The mass number of a nucleus is
- (a) Always less than its atomic number
 - (b) Always more than its atomic number
 - (c) Always equal to its atomic number
 - (d) Sometimes more than and sometimes equal to its atomic number
- Q2. Nuclear binding energy is equivalent to
- (a) Mass of proton
 - (b) Mass of neutron
 - (c) Mass of nucleus
 - (d) Mass defect of nucleus
- Q3. In nuclear reaction, there is conservation of
- (a) Mass only (b) Momentum only (c) Energy only (d) Mass, energy and momentum
- Q4. Particles which can be added to the nucleus of an atom without changing its chemical properties are called
- (a) Neutrons (b) Electrons (c) Protons (d) Alpha particles
- Q5. The radius of a nucleus is
- (a) Directly proportional to its mass number
 - (b) Inversely proportional to its atomic weight
 - (c) Directly proportional to the cube root of its mass number
 - (d) None of these
- Q6. The mass of an atomic nucleus is less than the sum of the masses of its constituents. This mass defect is converted into
- (a) Heat energy (b) Light energy (c) Electrical energy (d) Energy which binds nucleons together
- Q7. The neutrons and protons are collectively called as
- a) Neutrons (b) Mass (c) Nucleons (d) None

ASSERTION AND REASON QUESTIONS

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
- (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
- (c) If assertion is true but reason is false.
- (d) If the assertion and reason both are false.
- (e) If assertion is false but reason is true

Q8. Assertion : Density of all the nuclei is same.

Reason : Radius of nucleus is directly proportional to the cube root of mass number

Q9. Assertion : All the radioactive elements are ultimately converted in lead.

Reason : All the elements above lead are unstable

Q10. Assertion : The mass of a nucleus can be either less than or more than the sum of the masses of nucleons present in it.

Reason : The whole mass of the atom is considered in the nucleus.

2 MARKS QUESTIONS

Q11. Why are heavy nuclei usually unstable?

Q12. Write two characteristic features of nuclear force which distinguish it from Coulomb's force

Q13. Why do stable nuclei never have more protons than neutrons?

Q14. Distinguish between isotopes and isobars. Give one example for each of the species.

3-MARKS QUESTIONS

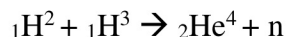
Q 15. Define nuclear fusion. Nuclear fusion is not possible in laboratory. Explain.

Q16. Calculate binding energy per nucleon of $^{56}\text{Fe}_{26}$ nucleus. Given that mass of $^{56}\text{Fe}_{26} = 55.934939\text{u}$, mass of proton = 1.007825u , mass of neutron = 1.008665u and $1\text{u} = 931\text{ MeV}$.

- 1) $^3\text{He}_2$ and $^3\text{H}_1$ nuclei have same mass number. Do they have the same Binding energy
Explain
- 2) If both the number of neutrons and the numbers of protons are conserved in each nuclear reaction, in what way is mass converted into energy in a nuclear reaction? Explain

5-MARKS QUESTIONS

Q 17. Distinguish between nuclear fission and fusion. Show how both these processes energy is released. Calculate the energy release in MeV in the deuterium-tritium fusion reaction



using the data mass of ${}_1\text{H}^2 = 2.014102 \text{ u}$, mass of ${}_1\text{H}^3 = 3.016949 \text{ u}$, mass of ${}_2\text{He}^4 = 4.002603 \text{ u}$, mass of neutron = 1.008665 u , $1 \text{ u} = 931.5 \text{ MeV}$

Q18. i) What characteristic property of nuclear force explains the constancy of binding energy per nucleon (B.E/A) in the range of mass number 'A' lying $30 < A < 170$. ii) Show that the density of nucleus over a wide range of nuclei is constant independent of mass number A.

3-MARKS QUESTIONS- CASE STUDY

Q19. As per the population rise in our country, energy demand is also increasing especially the electrical energy. For fulfilment of such demand, one of the option is to utilize nuclear sources. Nuclear energy is obtained through the nuclear fission process, where a bi nucleus gets split into two or more than two smaller nuclei along with tremendous amount of energy. This energy can be used for either constructive purpose or destructive purpose for the humans. Nuclear reaction is of two types- nuclear fission and nuclear fusion. In nuclear reactor, controlled nuclear fission reaction is carried out.

(I) The process taking out in the sun, due which we get light and heat energy

(A) Nuclear fission (B) Nuclear fusion (C) Thermal reactions (D) Nuclear holocaust

(II) The atomic energy programme in our country was launched under the leadership of

(A) C V Raman (B) H J Bhabha (C) A P J Abdul Kalam (D) Amit Bhatnagar

(III) The temperature of the core of the sun is about

(A) 10^3 to 10^4 K (B) 10^6 to 10^7 K (C) 10^9 to 10^{10} K (D) None of the above

(IV) To slow down the fast moving neutrons in nuclear reactor by

(A) Control rods (B) Moderator (C) Coolant (D) Plasamon.

Q 20. *Read the passage given below and answer the following questions:*

Neutrons and protons are identical particle in the sense that their masses are nearly the same and the force, called nuclear force, does into distinguish them. Nuclear force is the strongest force. Stability of nucleus is determined by the neutron proton ratio or mass defect or packing fraction. Shape of nucleus is calculated by quadrupole moment and spin of nucleus depends on even and odd mass number. Volume of nucleus depends on the mass number. Whole mass of the atom (nearly 99%) is centred at the nucleus

(I) The correct statements about the nuclear force is/are

- (a) charge independent
- (b) short range force
- (c) non-conservative force
- (d) all of these.

(II) The range of nuclear force is the order of

- (a) 2×10^{-10} m
- (b) 1.5×10^{-20} m
- (c) 1.2×10^{-4} m
- (d) 1.4×10^{-15} m

(III) A force between two protons is same as the force between proton and neutron. The nature of the force is

- (a)Electrical force
- (b) weak nuclear force
- (c) gravitational force
- (d) strong nuclear force

(IV) All the nucleons in an atom are held by

- (a) nuclear forces
- (b) vander waal's forces
- (c) tensor forces
- (d) coulomb forces

ANSWER KEY-SECTION A

MCQ: 1) b 2) d 3) b 4) a 5) c 6) d 7) c

ASSERTION AND REASON: 8) a 9) a 10) e

2-MARKS QUESTIONS

11. Due to large repulsive force between the large numbers of protons in the nucleus

12. Characteristic Features of Nuclear Force

Nuclear forces are short range attractive forces (range 2 to 3 fm) while Coulomb's forces have range up to infinity and may be attractive or repulsive.

Nuclear forces are charge independent forces; while Coulomb's force acts only between charged particle

13. Protons are positively charged and repel one another electrically. This repulsion becomes so great in nuclei with more than 10 protons or so, that an excess of neutrons which produce only attractive forces, is required for stability

14. Atoms of same elements having same atomic number but different mass numbers known as isotopes. Eg. ${}_{17}\text{Cl}^{35}$, ${}_{17}\text{Cl}^{36}$ and ${}_{17}\text{Cl}^{37}$

Atoms of different elements with different atomic number but same mass number known as isobars. Eg ${}_{18}\text{Ar}^{40}$, ${}_{20}\text{Ca}^{40}$

3-MARKS QUESTIONS

15. In nuclear fusion reaction, two or more than two smaller nuclei get combine together to more stable nucleus. Nuclear fusion is not possible in laboratory because it requires very high temperature such as 10^6 to 10^7K . Such high temperature are often generated in nuclear fission. That is why fission precedes fusion. These processes cannot be carried out in laboratory

16. Mass of proton = 1.007825 u

Mass of neutron = 1.008665 u

Mass of Fe = 55.934939 u

Mass of 26 protons = $26 \times 1.007825 \text{ u} = 26.2035 \text{ u}$

No of neutrons = $56 - 26 = 30$

Mass of 30 neutrons = $30 \times 1.008665 \text{ u} = 30.25995 \text{ u}$

Total mass of 56 nucleons = $26.2035 + 30.25995 = 56.4334 \text{ u}$

Mass defect = Total mass – mass of Fe = 56.4334 – 55.934939 = 0.528468 u

Binding energy = mass defect x 931.5 MeV = 0.528468 x 931.5 MeV

= 492.267942 MeV

Binding energy per nucleon = Binding energy/ mass number = 492.267942/56

= 8.790 MeV

- 1) No, the binding energy of ${}^3_1\text{H}$ is greater since more energy is required to bind 2 neutrons and 1 proton in ${}^3_1\text{H}$ than 2 protons and 1 neutron in ${}^3_2\text{He}$
- 2) As the number of nucleons is conserved in a nuclear reaction, the total rest mass of protons and neutrons on each other side of the reaction remains same. But the binding energies of nuclei on the two sides of the reaction different. It is this difference in B.E that appears as the energy released in the nuclear reactions.

5 MARKS QUESTIONS

17.

Nuclear Fission	Nuclear Fusion
When the nucleus of an atom splits into lighter nuclei through a nuclear reaction the process is termed nuclear fission.	Nuclear fusion is a reaction through which two or more light nuclei collide with each other to form a heavier nucleus.
When each atom split, a tremendous amount of energy is released	The energy released during nuclear fusion is several times greater than the energy released during nuclear fission.
Fission reactions do not occur in nature naturally	Fusion reactions occur in stars and the sun
Little energy is needed to split an atom in a fission reaction	High energy is needed to bring fuse two or more atoms together in a fusion reaction
Atomic bomb works on the principle of nuclear fission	Hydrogen bomb works on the principle of a nuclear fusion bomb.

In both reactions there is a mass defect which is converted into energy.

Now energy released in the reaction

$$\begin{aligned}
 Q &= [\text{mass}({}_1^2\text{H}) + \text{mass}({}_1^3\text{H}) - \text{mass}({}_2^4\text{He}) - \text{mass of neutron}]c^2 \\
 &= [2.014102 + 3.016049 - 4.002603 - 1.008665] \times 931.5 \text{ MeV} \\
 &= 0.018883 \times 931.5 \text{ MeV} = 17.58 \text{ MeV}
 \end{aligned}$$

18.i) The approximate constancy of BE/A over most of the range is saturation property of nuclear force. In heavy nuclei, nuclear size > range of the nuclear force.

So a nucleus contains approximately a constant number of neighbors and thus the nuclear BE/A levels off at high A. This saturation of the nuclear force

ii) The mass of the nucleus of the atom of the mass number A

$A = A \text{ a.m.u.}$

$$= A \times 1.660565 \times 10^{-27} \text{ kg}$$

The volume of the nucleus is $\frac{4}{3}\pi r^3 = \frac{4}{3}\pi(R_0 A^{1/3})^3 = \frac{4}{3}\pi R_0^3 A$

$$\therefore \text{Volume} = \frac{4}{3}\pi (1.1 \times 10^{-15})^3 \times A \text{ m}^3$$

Now the density of the nucleus = m/V ; where m is mass and V is the volume of the nucleus.

$$\rho = A \times 1.660565 \times 10^{-27} \text{ kg} / \frac{4}{3}\pi (1.1 \times 10^{-15})^3 = 2.97 \times 10^{17} \text{ Kg m}^{-3}$$

Thus, we can see the density of the nuclei is independent of mass number and it is constant for all nuclei

3 MARKS QUESTIONS- CASE STUDY

19. (I) b (II) b (III) b (IV) a

20. (I) a (II) d (III) d (IV) a

SECTION-B

MCQ

Q1. Binding energy per nucleon of a stable nucleus is

- (a) 8 eV (b) 8 KeV (c) 8 MeV (d) 8 BeV

Q2. The binding energy per nucleon is almost constant for many nuclei. It shows that nuclear forces are

- (a) Charge independent
(b) Saturated in nature
(c) Short range in nature
(d) Attractive in nature

Q3. The electrons cannot exist inside the nucleus because

- (a) de-Broglie wavelength associated with electron in β -decay is much less than the size of nucleus
(b) de-Broglie wavelength associated with electron in β -decay is much greater than the size of nucleus
(c) de-Broglie wavelength associated with electron in β -decay is equal to the size of nucleus
(d) Negative charge cannot exist in the nucleus

Q4. Which one of the following has the identical property for isotopes?

- (a) Physical property (b) Chemical property (c) Nuclear property (d) Thermal property

Q5. When the number of nucleons in nuclei increases, the Binding energy per nucleon

- (a) Increases continuously with mass number
(b) Decreases continuously with mass number
(c) Remains constant with mass number
(d) First increases and then decreases with increase of mass number

Q6. Fusion reactions take place at high temperature because

- (a) Atoms are ionised at high temperature
(b) Molecules break up at high temperature
(c) Nuclei break up at high temperature
(d) Kinetic energy is high enough to overcome repulsion

Q7. The tritium which is the isotope of hydrogen contains

- (a) One proton, one neutrons
(b) One proton, two neutrons

(c) Two protons, one neutrons

(d) None

Q8. Heavy stable nuclei have more neutrons than protons. This is because of the fact that

(a) Neutrons are heavier than protons.

(b) Electrostatic force between protons is repulsive.

(c) Neutrons decay into protons through beta decay.

(d) Nuclear forces between neutrons are weaker than that between protons

ASSERTION AND REASON QUESTIONS

Read the assertion and reason carefully to mark the correct option out of the options given below:

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(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(c) If assertion is true but reason is false.

(d) If the assertion and reason both are false.

(e) If assertion is false but reason is true

Q9. Assertion : Fragments produced in the fission of ^{235}U are radioactive.

Reason : The fragments have abnormally high proton to neutron ratio.

Q10. Assertion : All the radioactive elements are ultimately converted in lead.

Reason : All the elements above lead are unstable.

Q11. Assertion : It is not possible to use $\text{Cl } 35$ as the fuel for fusion energy.

Reason : The binding energy of $\text{Cl } 35$ is too small.

Q12. Assertion : Electron capture occurs more often than positron emission in heavy elements.

Reason : Heavy elements exhibit radioactivity.

(2 MARKS QUESTIONS)

Q13. For a given nuclear reaction the B.E./nucleon of the product nucleus/nuclei is more than that for the original nucleus/nuclei. Is this nuclear reaction exothermic or endothermic in nature? Justify your choice

Q14. Calculate the energy equivalent of 1amu in MeV .

(3 MARKS QUESTIONS)

Q15. A given coin has a mass of 3.0 g. Calculate the nuclear energy that would be required to separate all the neutrons and protons from each other. For simplicity assume that the coin is entirely made of $^{63}_{29}\text{Cu}$ atoms (of mass 62.92960 u)

Q16. Calculate binding energy per nucleon of $^{209}_{83}\text{Bi}$ nucleus. Given that mass of $^{209}_{83}\text{Bi} = 55.934939\text{u}$, mass of proton = 1.007825u, mass of neutron = 1.008665 u and $1\text{ u} = 931\text{ MeV}$.

Q17. Suppose, we think of fission of a $^{56}_{26}\text{Fe}$ nucleus into two equal fragments, $^{28}_{13}\text{Al}$. Is the fission energetically possible? Mass of $^{56}_{26}\text{Fe} = 55.934939\text{u}$, mass of $^{28}_{13}\text{Al} = 55.934939\text{u}$

- 1) How is the size of a nucleus experimentally determined? Write the relation between the radius and mass number of the nucleus is independent of its mass number

ANSWER KEY-SECTION B

MCQ: 1. C 2. C 3. B 4. B 5. D 6. A 7. 8. B

ASSERTION AND REASON: 9. C 10. C 11. C 12. B

2 MARKS QUESTIONS

13. As BE/nucleon of the product nuclei is more than that of original nuclei. So more mass has been converted into energy. This would result in release of energy so it is exothermic

14. One atomic mass unit (1 a.m.u.) is defined as one twelfth of the mass of an atom of carbon-12.

$$1 \text{ a.m.u.} = 1.66 \times 10^{-27} \text{ kg}$$

According to Einstein's mass-energy equivalence relation,

$$E = (\Delta m)c^2$$

So, we need to find energy equivalent of 1 a.m.u.

$$E = (1.66 \times 10^{-27} \text{ kg}) \times (3 \times 10^8 \text{ m/s})^2$$

$$= 1.49 \times 10^{-10} \text{ joules}$$

$$\text{But } 1 \text{ MeV} = 1.6 \times 10^{-13} \text{ J}$$

$$E = 1.49 \times 10^{-10} \times 1.6 \times 10^{-13} \text{ MeV}$$

$$= 931.5 \text{ MeV}$$

Thus, 1 amu of mass is equivalent to 931.5 MeV of energy.

3-MARKS QUESTIONS

$$15. \text{Number of atoms in a 3g coin} = 6.023 \times 10^{23} \times 3 / 63 = 2.868 \times 10^{22}$$

Each copper atom has 29 protons and 34 neutrons.

$$\text{Mass of the copper nucleus} = 62.92960$$

$$\text{Thus, the mass defect of each atom} = 29 \times 1.00783 + 34 \times 1.00867 - 62.92960 = 0.59225 \text{ u.}$$

$$\text{Total mass defect of all atoms} = 0.59225 \text{ u} \times 2.868 \times 10^{22} = 1.6985 \times 10^{22} \text{ u.}$$

$$\text{Thus, the nuclear energy required} = 1.6985 \times 10^{22} \times 931 \text{ MeV} = 1.58 \times 10^{25} \text{ MeV.}$$

16. Bismuth nucleus contains 83 protons

Number of neutrons = $209 - 83 = 126$ neutrons

Now, Mass of 83 protons = $83 \times 1.007825 = 83.649475 \text{ amu}$

Mass of 126 neutrons = $126 \times 1.008665 = 127.091790 \text{ amu}$

Therefore, total mass of nucleons = $83.649475 + 127.091790 = 210.741260 \text{ amu}$

Given, mass of nucleus = 208.980388 amu

Now, mass defect $\Delta m = 210.741260 - 208.980388 = 1.760872$

Total binding energy = $1.760872 \times 931.5 = 1640.26 \text{ MeV}$

Therefore, average binding energy per nucleon = $1640.26 / 209 = 7.848 \text{ MeV}$

17. It is given that The atomic mass of Fe = $55.93494.0 \text{ u}$

The atomic mass of Al = 27.98191 u

The Q value of this nuclear reaction is given as $Q = [(\text{mass of Fe}) - (2 \text{ mass of Al})] c^2$
 $= [55.93494 - 2 \times 27.98191] c^2$

$Q = (-0.02888 c^2) \text{ u}$

But we know that $1 \text{ u} = 931.5 \text{ MeV}/c^2$ $Q = -0.02888 \times 931.5 = -26.902 \text{ MeV}$

The Q value of the fission is negative. Therefore, fission is not possible. For a possible fission reaction, the Q value should be positive

- 1) It was possible to obtain the size of the nucleus through Rutherford's experiment. We can calculate the size of the nucleus, by obtaining the point of closest approach of an alpha particle

If R is the radius of nucleus having mass number A, then $R = R_0 A^{1/3}$

Where $R_0 = 1.2 \times 10^{-15} \text{ m}$ which is the range of order of nuclear size and A is mass number

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SECTION-C

(3 MARKS QUESTIONS)

Q1. A heavy nucleus X of mass number $A = 240$ and $B.E/A = 7.6 \text{ MeV}$ is split into two nearly equal fragments Y and Z of mass numbers $A_1 = 110$ and $A_2 = 130$. The binding energy of each one of these nuclei is 8.5 MeV per nucleon. Calculate the total binding energy of each one of the nuclei X, Y and Z and hence the energy Q released per fission in MeV

Q2. Four nuclei of an element fuse together to form a heavier nucleus. If the process is accompanied by release of energy, which of the two the parent or the daughter nucleus would have higher binding energy

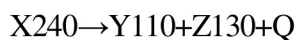
(2 MARKS QUESTIONS)

Q3. In tropical nuclear reaction e.g., ${}^2\text{H}_1 + {}^2\text{H}_1 \rightarrow {}^3\text{He}_2 + {}^1_0\text{n} + 3.27 \text{ MeV}$, although number of nucleons is conserved, yet energy is released. How? Explain

Q4. Obtain approximately the ratio of the nuclear radii of the gold isotope ${}_{79}\text{Au}^{197}$ and the silver isotope ${}_{47}\text{Ag}^{107}$

ANSWER KEY-SECTION C

1. The nuclear reaction is



As per given data

$$\begin{aligned} Q &= 110 \times 8.5 + 130 \times 8.5 - 240 \times 7.6 \\ &= 240(0.9) \text{ MeV} = 216 \text{ MeV} \end{aligned}$$

2. Since the unstable parent nuclei fuse to form a heavier stable daughter nuclei in a nuclear fusion reaction releasing some energy. So, daughter nuclei is more stable than parent nuclei. Thus daughter nucleus has more binding energy per nucleon than parent nucleus

3. In a nuclear reaction, the aggregate of the masses of the target nucleus (${}_1\text{H}^2$) and the bombarding particle may be greater or less than the aggregate of the masses of the product nucleus (${}_2\text{He}^3$) and the outgoing particle (${}_0\text{n}^1$).

So, from the law of conservation of mass-energy some energy (3.27 MeV) is evolved or involved in a nuclear reaction. This energy is called Q-value of the nuclear reaction.

4. Let R_{Au} be the nuclear radius of the gold isotope and R_{Ag} be the nuclear radius of the silver isotope.

We have the mass number of the gold (Au) is 197

the mass number of the silver (Ag) is 107.

The ratio of the radii of the two nuclei is related with their mass number as:

$$(R_{\text{Au}} / R_{\text{Ag}}) = (197/107)^{1/3} = 1.2256 = 1.23$$
